# Lecture 1: Motivation and AI History

Viliam Lisý & Branislav Bošanský

Artificial Intelligence Center
Department of Computer Science, Faculty of Electrical Eng.
Czech Technical University in Prague

viliam.lisy@fel.cvut.cz

February, 2026

# Lecturers



Branislav Bošanský
Associate professor at CTU
Research manager at Gen (Avast)
Postdoc from Aarhus University, Denmark
50+ publications in top AI venues



Viliam Lisý
Associate professor at CTU
Ex AI architect at Gen
Postdoc from University of Alberta, Canada
Co-author of DeepStack poker AI
50+ publications in top AI venues

# What is AI?

Britannica: Artificial intelligence (AI), the ability of a digital computer or computer-controlled robot to perform tasks commonly associated with intelligent beings.

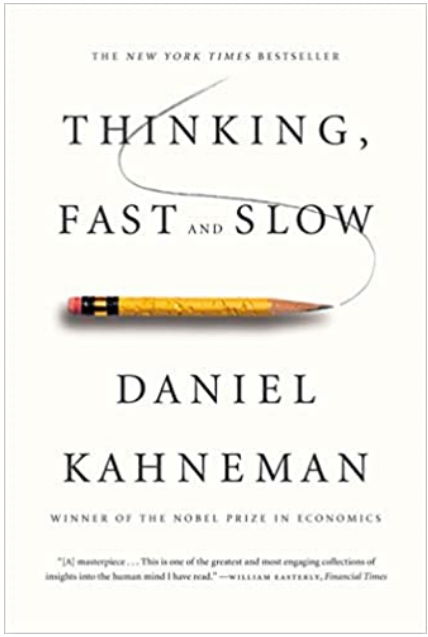
Merriam-Webster:

1. a branch of computer science dealing with the simulation of intelligent behavior in computers
2. the capability of a machine to imitate intelligent human behavior

For this lecture: An AI system chooses actions (or outputs) that maximize (expected) success given (incomplete) information.

# Thinking Fast and Slow

Humans have two main types of thinking: "thinking fast" — reflexive, without much introspection — and "thinking slow" — based on internal models and predictions about the world.




My analogy:

> Machine Learning $\approx$ thinking fast (System 1)
>
> Artificial Intelligence $\supset$ thinking slow (System 2)

Modern breakthroughs **combine both**. This course focuses on System 2: search, planning, reasoning — but we will show how System 1 (learned functions) lets these methods **scale**.

# The Power of Abstraction

These look like completely different problems:

1. A robot navigating a building
2. Managing inventory in a warehouse
3. Playing Sokoban (single-player computer game)
4. Choosing a medical treatment plan

But they all share the same abstract structure: **Markov Decision Process** (MDP).

**Key idea of this course**

Learn to recognize the abstract model behind a real-world problem $\Rightarrow$ get algorithms for free.

# Goals of the Course

Understand the problems and methods in the field of AI.

Two recurring themes:

1. **Abstraction**: real-world problems $\rightarrow$ formal models $\rightarrow$ general algorithms

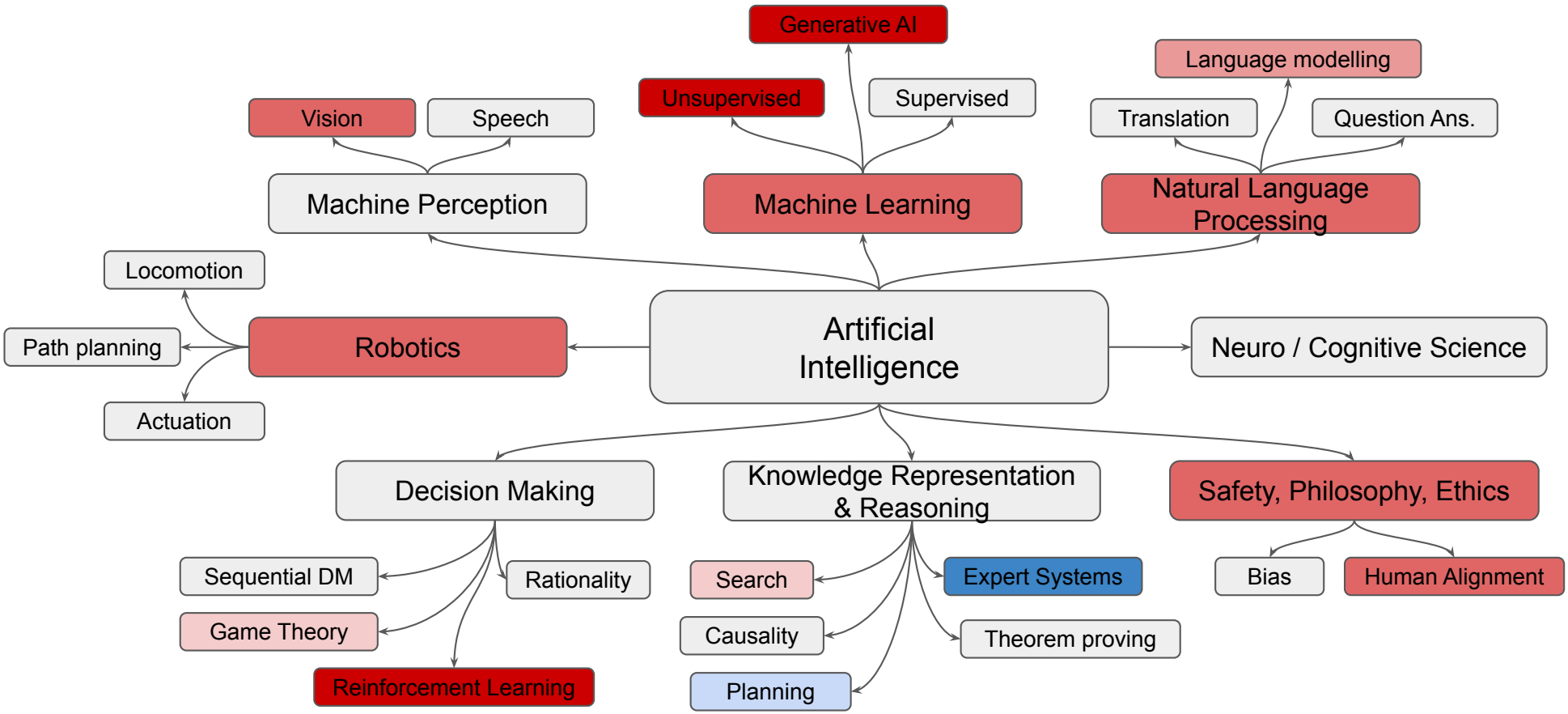
   MDP, state-space search, CSP, Bayesian networks, ...

2. **Scaling via function approximation**: neural nets let you scale the components of these abstract models to high-dimensional, real-world settings

Get broader context of the field

- Understand the scope of AI vs. ML vs. CS
- Understand the hypes and limitations, PoC vs. product

# AI Topics Overview



# Structure of Today's Lecture

1. What is AI, two themes of the course
2. **Supervised learning as a black box** (new this year)
   - what it does, how to use it
   - why it matters for every topic in this course
3. AI history through three threads
   - robotics
   - decision making
   - natural language processing
4. Course organisation

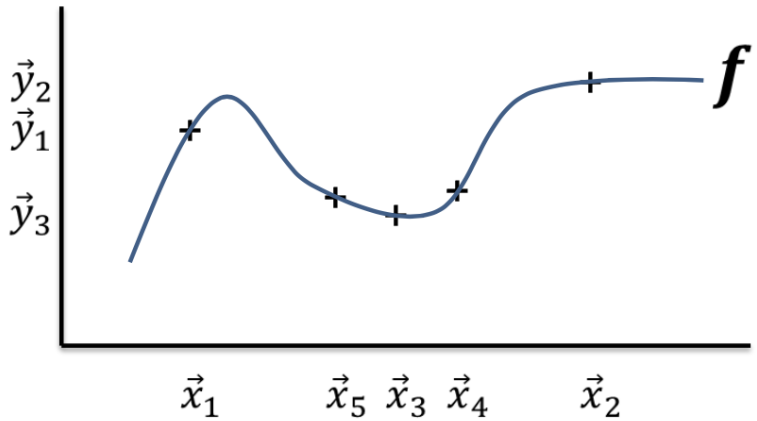
# Supervised Learning = Function Fitting

A powerful tool for AI (details in ML courses, here just the **black box** view).

Supervised learning = fitting a (high dimensional) function

For a data set $(\vec{x_i}, \vec{y_i})$, find a function $f$ that minimizes:

$$\frac{1}{n}\sum_i \|f(\vec{x_i}) - \vec{y_i}\|.$$




For example, $f($  $) = 2, f($  $) = 3, f($  $) = 4, f($  $) = 5.$

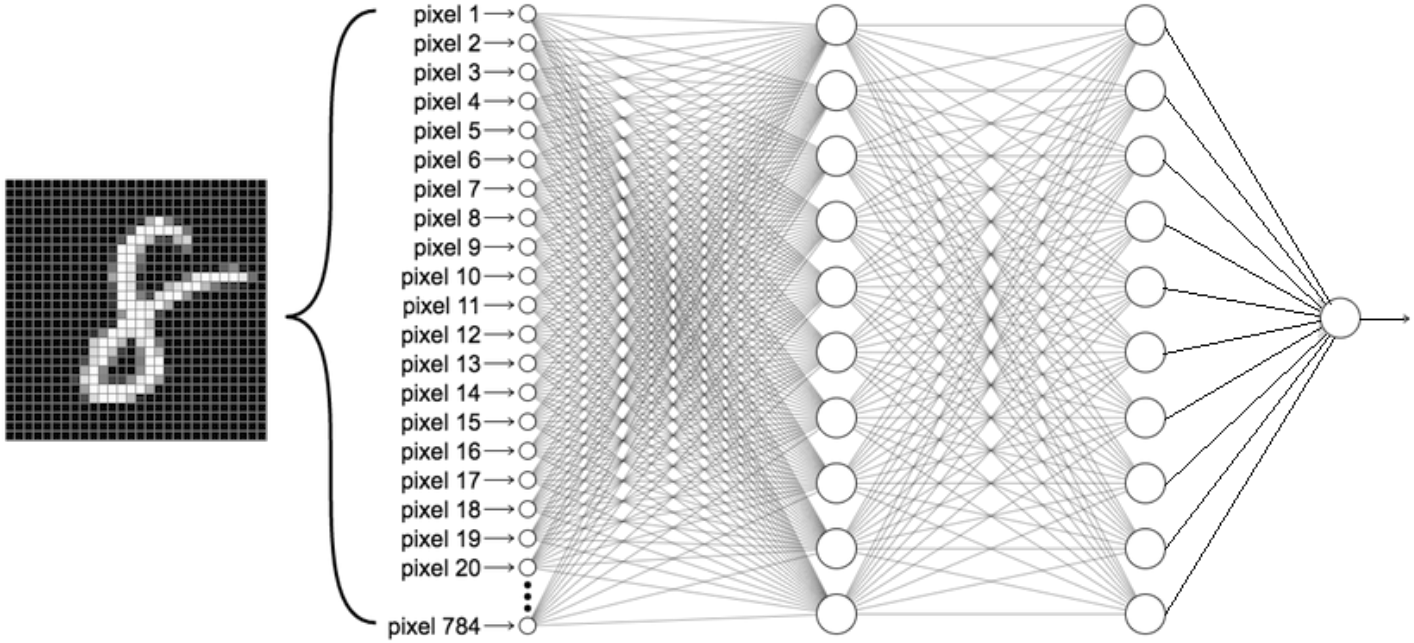
# Examples of Learned Functions

The same idea works at very different scales:

| **Input** $\vec{x}$ | **Output** $\vec{y}$ | **Application** |
|---|---|---|
| 1D number | 1D number | curve fitting |
| $28 \times 28$ pixel image | digit 0–9 | handwriting (MNIST) |
| $224 \times 224$ image | 1 of 1000 classes | ImageNet classification |
| text (sentence) | positive/negative | sentiment analysis |
| English sentence | Czech sentence | machine translation |
| game state image | action to take | game playing (Atari) |

Key message: given enough data and compute, neural nets can approximate **surprisingly complex mappings**.

# Neural Network — A Parametric Function




- A neural network is just a parametric function $f_{\vec{w}}(\vec{x})$
- Parameters $\vec{w}$ ("weights") are adjusted during training
- RELU unit: $y = \max\{0, \sum_{i=0}^{n} w_i x_i\}$
- You do not need to understand the training details for this course

# How to Use It in Practice

The modern workflow:

1. **Collect data**: pairs $(\vec{x_i}, \vec{y_i})$
2. **Choose architecture**: e.g., convolutional network for images, transformer for text
3. **Choose loss**: what is a better (imperfect) fit, e.g., MSE
4. **Train**: framework (PyTorch, JAX) does the heavy lifting
5. **Evaluate**: test on *held-out data* the model has never seen

The one thing you **must** understand:

**Generalization**

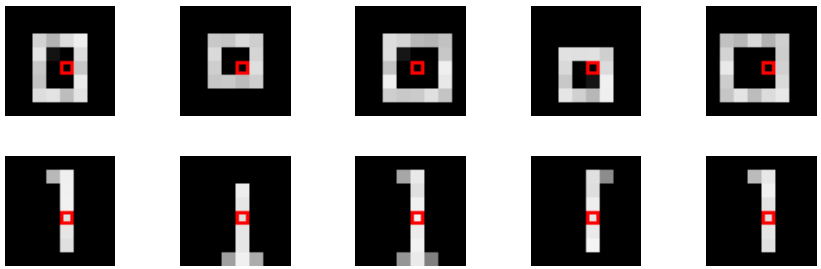
A model that memorises training data but fails on new data is often useless.
Always measure performance on a **separate test set**.
When training error $\ll$ test error, the model is **overfitting**.

# Overfitting — Why Generalization Matters

**Training data** — look at the center pixel:

**Zeros:**

**Ones:**

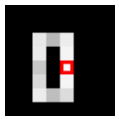


## A "lazy" model

The center pixel is **dark** for every 0 and **bright** for every 1.
A model using only this pixel achieves **training accuracy = 100%**.

Like a student who memorises answers instead of understanding the material.

# Overfitting — But on New Data...



Narrow zero (center bright)

Model says: "one" ✗



Shifted one (center dark)

Model says: "zero" ✗

The single-pixel rule fails on both. **Test accuracy = 0%.**
The model memorised an *accident* of the training data.

## The real problem

Not just that the classifier fails — we **have no way of knowing** how bad it is after training.
100% training accuracy tells us *nothing* about real-world performance.
**Always evaluate on held-out data.**

# Why This Matters for This Course

Abstraction gives you the **algorithm**. Function approximation lets you **scale** its components.

| **Abstract model** | **Component needed** | **Scaling via learning** |
|---|---|---|
| A* search | heuristic $h(s)$ | hand-crafted for small problems, **learned** for large state spaces |
| MDP / RL | value function $V(s)$ | tabular for small MDPs, **learned** for Atari (DQN) |
| Game trees | evaluation function | hand-tuned for Deep Blue, **learned** for AlphaGo |
| Bayes nets / POMDPs | observation model | specified by experts for simple models, **learned** from data at scale |

## Function approximation

Throughout this course, we will define abstract models and their algorithms. Then we will show how function approximation lets those same algorithms scale to real-world complexity.

# Why is it Important to Know AI History

1. what is already possible / known and what is not
2. when did the well-known results happen
3. hype cycles and actual progress
4. time from flashy prototypes in media to practice
5. not to be embarrassed by lack of general knowledge
    - Turing test, Deep Blue, AI winter, superintelligence, etc.

# $-\infty$ – 1940: “Ancient” history

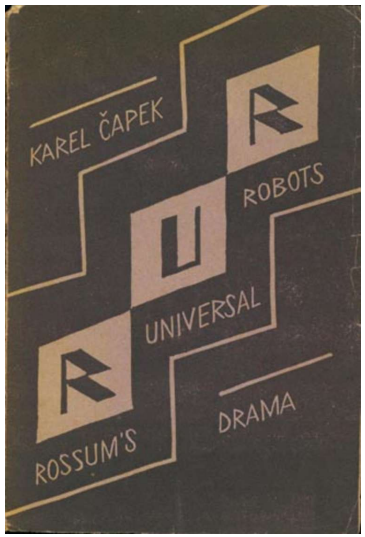
long BC: Golem

1914 First actual chess-playing machine

Endgame of rook+king vs. king

1921 RUR

Origin of the word “robot”

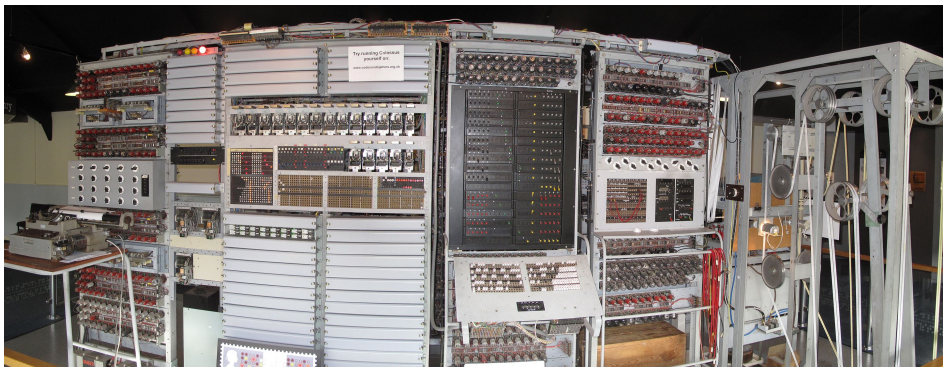


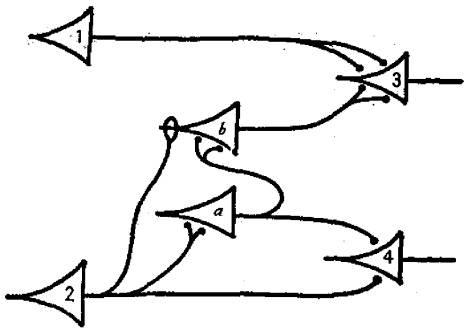
1941 First electronic computers

Z3, Atanasoff–Berry, Colossus

1943 McCulloch&Pitts create the first model of artificial neurons

1945 Alan Turing states that computers could play chess





# 1950: Turing Test

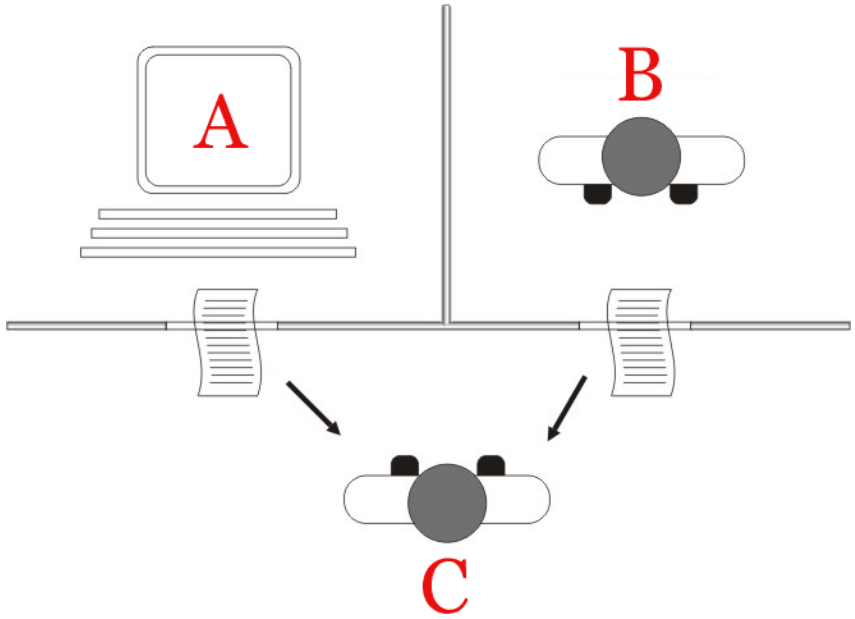
Formalization of the question: **Can machines think?**

The first variant was proposed in 1948 using Chess. Can a computer be programmed to be indistinguishable from a bad chess player?

## Official publication

A. M. Turing (1950): Computing Machinery and Intelligence. MIND a Quaterly Review of Psychology and Philosophy, Vol. 59, No. 236., pp 433–460.

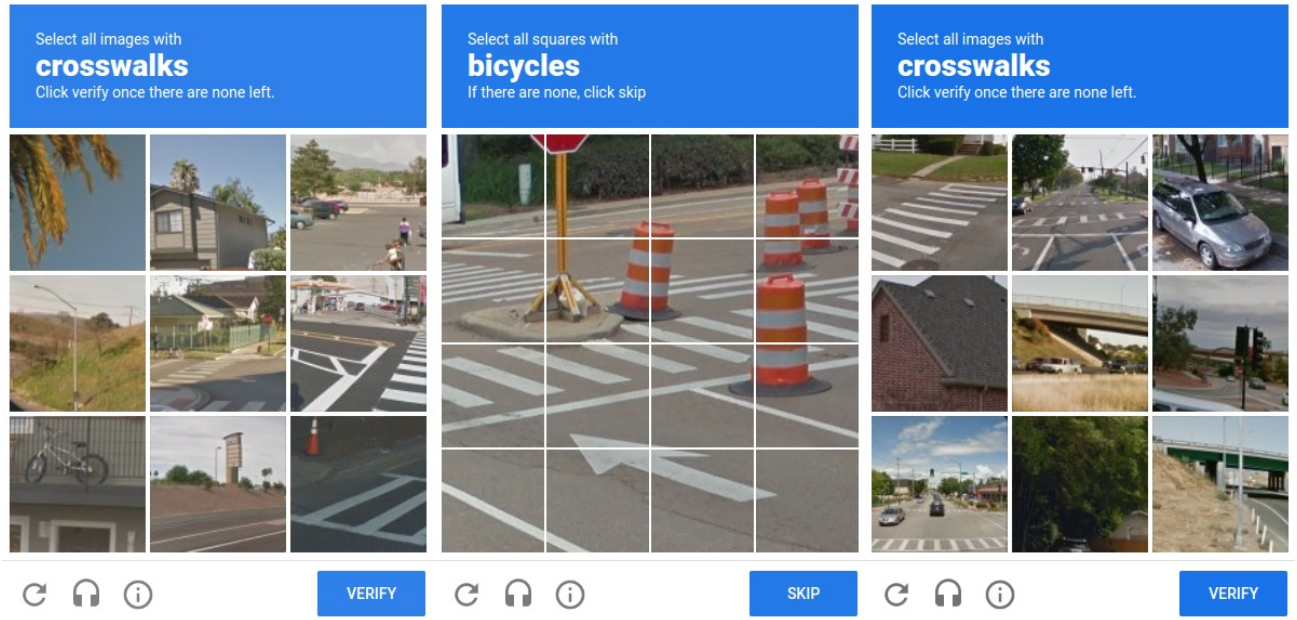
# 1950: Turing Test



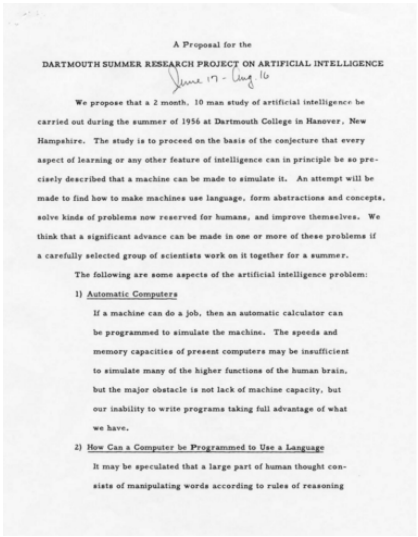

- Separates physical and intellectual capabilities
- Turing proposed variants
  - An interrogator distinguishes between a human and a machine
  - She distinguishes whether the subject is a human or a machine
  - (1952) A whole jury serves as the interrogator
- The paper rebuts some of the main objections

# 2026: Turing Test

CAPTCHA: Completely Automatic Public Turing tests to tell Computers and Humans Apart



# 1956: Dartmouth Workshop on "Artificial Intelligence"



A Proposal for the

DARTMOUTH SUMMER RESEARCH PROJECT ON ARTIFICIAL INTELLIGENCE

June 17 - Aug 16

We propose that a 2 month, 10 man study of artificial intelligence be carried out during the summer of 1956 at Dartmouth College in Hanover, New Hampshire. The study is to proceed on the basis of the conjecture that every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it. An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves. We think that a significant advance can be made in one or more of these problems if a carefully selected group of scientists work on it together for a summer.

The following are some aspects of the artificial intelligence problem:

1) Automatic Computers

If a machine can do a job, then an automatic calculator can be programmed to simulate the machine. The speeds and memory capacities of present computers may be insufficient to simulate many of the higher functions of the human brain, but the major obstacle is not lack of machine capacity, but our inability to write programs taking full advantage of what we have.

2) How Can a Computer be Programmed to Use a Language

It may be speculated that a large part of human thought consists of manipulating words according to rules of reasoning



The proposal from 1955 used the term "artificial intelligence".

- "We propose that a 2 month, 10 man study of artificial"...
- Main topics:
  - Automatic computers
  - How can a computer be programmed to use a language
  - Neuron nets
  - Theory of the size of a calculation
  - Self-improvement
  - Abstractions
  - Randomness and Creativity

Considered to be the founding event of AI as a research field.

# Robotics

Mechanical eng. + electrical eng. + sw eng. + control eng. + **AI**

Creating intelligent behavior of machines in the **physical space**

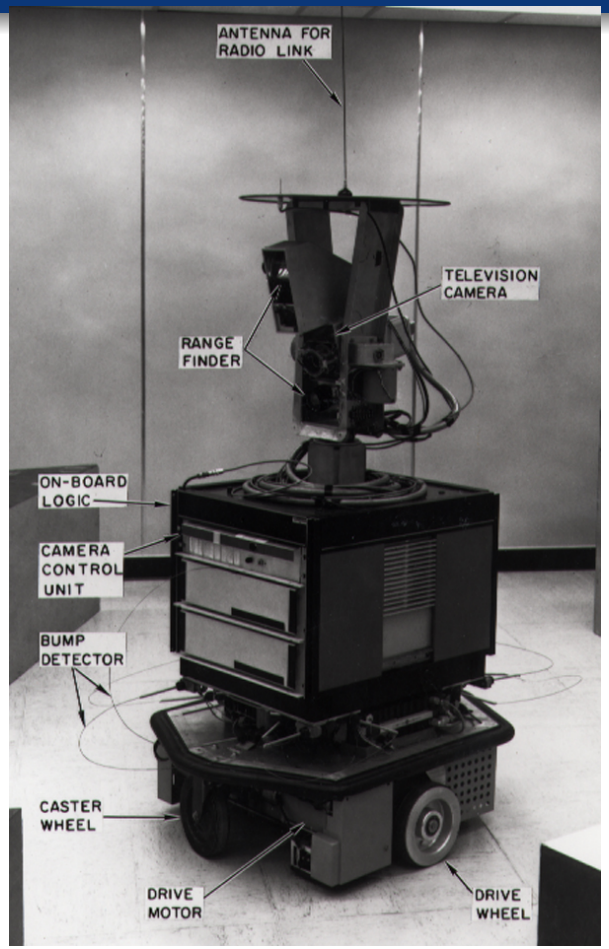
- Sensing the environment: bumpers, odometers, LIDARs, cameras, object recognition
- Thinking: mapping, path/movement planning, task planning
- Acting: movement, grasping, speaking

Are we missing anything? How do we know?

# Robotics — Shakey (1966–1972)

- 1961: Unimate — the first industrial robot (video)
- 1966–1972: Shakey — universal platform at Stanford Research Institute

  Combined navigation, computer vision, natural language processing, etc.
- Programmed primarily in **Lisp**. General purpose planning using **STRIPS**. The project also introduced the **A*** algorithm.




## Abstraction

Shakey's team *abstracted* robot planning as **state-space search** ⇒ invented A* and STRIPS. Those same abstractions are inside every modern robot.

# Robotics — Self-Driving

- 1986: VaMoRs — Driverless van
  - Drives 55 mph on empty streets of Bavaria
  - Driving in public traffic since 1992, 1000s kms on highways








- 2005: DARPA Grand Challenge

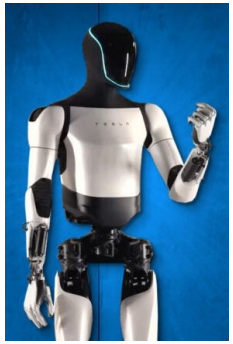
- 2024–2026: Waymo — over 20 million rides completed, expanding to 20 more cities



# Robotics Now

Humanoids





Drones and delivery





Even the most advanced robots combine:

- **learned** perception and control policies (System 1)
- A*, planning, RL **algorithms** underneath (System 2)

# Robotics — Lessons Learned

- Progress in robotics is incredibly slow
- Best way to understand complex systems is to build them
- Flashy media results are not very representative
- Competitions are a good driver of actual progress in AI
- Shakey's team *abstracted* planning as search $\Rightarrow$ same abstractions used 60 years later
- What changed: modern robots *learn* perception and control, so the abstract algorithms work in the real world

# AI Decision Making

Choosing the right action based on available information

- next move in a game based on rules
- right diagnosis based on symptoms
- next task to achieve a complex goal

# Decision Making I — Rules and Expert Systems

1950 Turing wrote the first chess program.

1950 Claude Shannon published the first paper on computer chess.

1956 Logic Theorist: Reasoning as heuristic search

> Eventually proved 38 of the first 52 theorems in chapter 2 of the Principia Mathematica

1966 Shakey the robot (A*, STRIPS)

1972 MYCIN, an **expert system** to identify the source of infection and recommend antibiotics

- Knowledge base of ∼600 rules
- Acceptability rating of 65% from a panel of specialists (humans achieved 42.5% – 62.5%)
  **AI adoption is not only about performance**

# 1974–1980: The First AI Winter

Large drop of funding and interest in AI

- Initial hype did not deliver impactful applications
- UK funding cut due to Lighthill report

  > "In no part of the field have the discoveries made so far produced the major impact that was then promised"

- DARPA focussed on "mission-oriented direct research"
- Negative results of Minsky & Papert on perceptrons decreased popularity of connectionism



# Decision Making II — From Rules to Search

1980 XCON, **expert system** selecting computer components

Saved millions of dollars annually in deployment

1981 $850 million Japanese "Fifth Generation Computer" project

Supercomputer focused on parallel logical programming for AI

1988 Judea Pearl invented Bayesian networks

Turing Award in 2011

1987 – ∼1993: The Second AI Winter

1997 DeepBlue beats Kasparov

Parallel Alpha-Beta **search**, evaluation function **hand-tuned** on grandmaster games, openings database, endgames

# Decision Making III — AI Algorithms + Deep Learning

2011 Watson

2015 DQN in Atari games

RL + learned value function (neural net)

2015 AlphaGo

MCTS + learned evaluation + learner policy

2017 DeepStack (Poker)

CFR + RL + learned value function

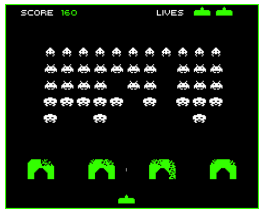
2019 AlphaStar (StarCraft)

Game Theory + RL + Deep Learning

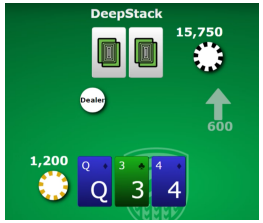
2022 Cicero (Diplomacy)

Tree search + RL + Deep Learning + LLMs








# Decision Making — Lessons Learned

- There are cycles of hype and slowdown
- Games (competitions) are a fundamental driver of progress
- Raw performance is not enough, there are trust and legal issues

**The pattern**

Deep Blue = game tree **search** + **hand-tuned** evaluation
AlphaGo = same abstract algorithm (MCTS) + **learned** evaluation + **learned** policy

The abstract algorithms (search, MCTS, RL) **stayed**. Function approximation allowed them to **scale**.

# Natural Language Processing

Understanding, interpreting, and generating human language.

- holding a conversation
- automated translation
- automated summarization
- question answering
- sentiment analysis

# Natural Language Processing — Eliza (1966)

Weizenbaum, J. (1966). ELIZA – a computer program for the study of natural language communication between man and machine. Communications of the ACM, 9(1), 36-45.

Shows that a computer can hold a natural language conversation.

Based on pattern matching with minimal permanent state.

- identifies keywords, discovers a minimal context, chooses a **transformation rule**, generic response if no keywords

Most famous set of conversation rules simulated a psychologist.

```
Men are all alike.
IN WHAT WAY?
They're always bugging us about something or other.
CAN YOU THINK OF A SPECIFIC EXAMPLE?
Well, my boyfriend made me come here.
YOUR BOYFRIEND MADE YOU COME HERE?
He says I'm depressed much of the time.
I AM SORRY TO HEAR YOU ARE DEPRESSED.
It's true. I am unhappy.
DO YOU THINK COMING HERE WILL HELP YOU NOT TO BE UNHAPPY.
I need some help, that much seems certain.
WHAT WOULD IT MEAN TO YOU IF YOU GOT SOME HELP?
Perhaps I could learn to get along with my mother.
TELL ME MORE ABOUT YOUR FAMILY.
My mother takes care of me.
WHO ELSE IN YOUR FAMILY TAKES CARE OF YOU?
My father.
YOUR FATHER?
You are like my father in some ways.
WHAT RESEMBLANCE DO YOU SEE?
You are not very aggressive
but I think you don't want me to notice that.
WHAT MAKES YOU THINK I AM NOT VERY AGGRESSIVE?
You don't argue with me.
WHY DO YOU THINK I DON'T ARGUE WITH YOU?
You are afraid of me.
DOES IT PLEASE YOU TO BELIEVE I AM AFRAID OF YOU?
My father is afraid of everybody.
WHAT ELSE COMES TO YOUR MIND WHEN YOU THINK OF YOUR FATHER?
Bullies.
DOES THAT HAVE ANYTHING TO DO WITH THE FACT
THAT YOUR BOYFRIEND MADE YOU COME HERE?
```

```
(0 YOU 0 ME) => (WHAT MAKES YOU THINK I 3 YOU)
```

# From Rules to Scale

1966 Eliza — simple rewriting rules

1995 A.L.I.C.E. — chatbot with complex pre-programmed rules

2006 Google Translate — statistical NLP

2011 Siri — still mostly rules

2013 Word2Vec (Mikolov et al. @ Google)

2018 BER**T** followed by other language models

2022 ChatGPT — massive pre-training and RLHF fine-tuning

2024 "Reasoning" models: o1, o3, R1

- Adding "System 2" to LLM inference
- Typically trained by **RL**

2025 LLM agents: LLMs calling tools, **planning**, writing code

- classical AI abstractions are coming back

# Language and AI — Lessons Learned

- Hand-coding knowledge did not make it
- Predicting the next token (word) and huge data did
- RL is used for LLM fine-tuning (RLHF)
- Recent models use "reasoning" related to **search** and **RL**
- LLM agents use **planning** and tool use — classical AI ideas

**Bridge to this course**

Even LLMs are now using search and planning. The abstract models you will learn in this course are more relevant than ever.

# The Bitter Lesson (Sutton, 2019)

Rich Sutton observed a recurring pattern across 70 years of AI research:

**The bitter lesson**

*"The biggest lesson that can be read from 70 years of AI research is that general methods that leverage computation are ultimately the most effective, and by a large margin."*

We saw this in all three threads:

- **Robotics**: hand-coded perception → learned perception and control
- **Decision making**: hand-tuned evaluation (Deep Blue) → learned evaluation + policy (AlphaGo)
- **NLP**: hand-crafted rules (Eliza) → next-token prediction at scale (GPT)

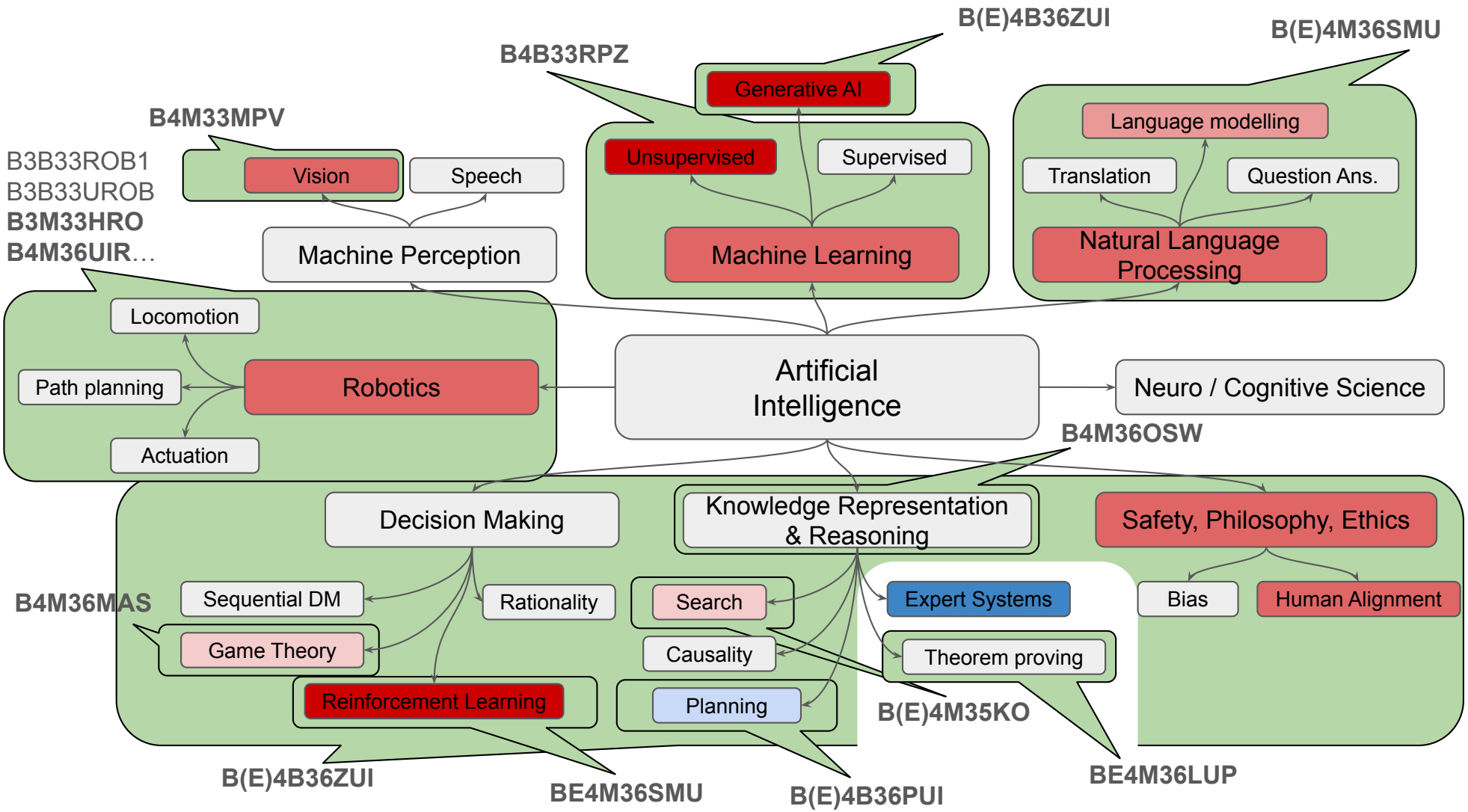
For this course: learn the **general abstract methods** (search, RL, MCTS, . . . ). They are the ones that survive — scaled by learned components.

# My Perspective

If I were to choose a field of study now, it would be AI!

- It is currently one of the most transformative technologies.
- AI is replacing jobs. Understanding it makes you the last to be replaced.
- Easy to find a job (anywhere).
- Interesting high-tech startup opportunities.

# AI Topics Overview



# Course Roadmap — Abstract Models

Lectures grouped by the **abstract model** you will learn:

| | |
|---|---|
| **State-space search** | L2: Formal representations & uninformed search<br>L3: Informed search, A* |
| **Structured states** | L4: CSP, Planning |
| **MDPs / RL** | L5–6: RL, solving MDPs<br>L7: Q-learning, DQN (+ function approx) |
| **Game trees** | L8: Two-player games (minimax, alpha-beta)<br>L9: MCTS, AlphaGo (+ function approx) |
| **Probabilistic models** | L10: Uncertainty in AI, Bayesian networks |
| **POMDPs** | L11: Sequential decisions with partial info. |
| **Generative AI** | L12: Introduction to generative AI |
| **AI Safety** | L13: AI safety, bias, and fairness |

# Course Organisation

https://cw.fel.cvut.cz/wiki/courses/zui/start

13 lectures leading towards some of the mentioned milestones




13 labs going deeper to selected algorithmic / theoretic topics








3 programming homeworks in python evaluated by BRUTE

Midterm test

Final written exam

# Course Grading

https://cw.fel.cvut.cz/wiki/courses/zui/start

30% for programming homework

- 10% State space search (A*) algorithm
- 10% Reinforcement learning
- 10% Game playing bot
- Extra tasks for additional points possible

Each task must be submitted for $\geq 50\%$ of its points

Deadline penalties: $\leq$ 24h: -20%; $>$ 24h: 0

Plagiarism will not be tolerated!

If you have serious issues **let us know** ASAP.

15% for the midterm written test

55% for the final written exam

in case of 80+% overall, also a brief oral exam

Standard evaluation scale: https://fel.cvut.cz/en/education/rules/Study_and_Exam_Code.pdf

# Course Literature

Slides are not study materials!

1. Take notes.
2. Artificial Intelligence: A Modern Approach (AIMA) by Stuart J. Russell and Peter Norvig (however, it is not free)
3. Reinforcement Learning: An Introduction by Richard S. Sutton and Andrew G. Barto (PDF available online)
4. Links on the courseware page and **in slides**
5. Wikipedia

